

**ON LICENSING BAREST CONDITIONALS:
FROM THE VANTAGE POINT OF THE SYNTAX-
PROSODY INTERFACE**

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ABSTRACT

This paper investigates two types of barest conditionals in Mandarin, both characterized by their minimal clausal structure and the presence of identical pairing, i.e., two duplicated verbs, such as *Gāi chī chī, gāi hē hē* ‘Eat when it's time to eat, drink when it's time to drink’, or *wh*-phrases, like *Ài shéi shéi* ‘Whoever it may be’ (cf. Tsai 2025). Based on the identical pairing analysis, this paper argues that both types are licensed by an explicit necessity operator, namely the sentence-initial modal: It unselectively binds the pair of identical variables, thereby licensing the donkey-conditional in question (cf. Cheng & Huang 1996, 2020). On the other hand, the type of identical pairing (i.e., verb vs. *wh*-phrase) depends on which variable acts as the information focus, while the sentence-initial modal serves to license the barest conditional construal. Furthermore, we argue that prosodic fusion, which merges the two separate conditional clauses into a single sentential intonation phrase (cf. Feng 2017), is key to identifying barest conditionals: Only after the fusion do the sentences need to observe the strict requirement of identical pairing. Finally, this paper brings to light the fascinating interaction among register, syntax, and prosody as observed in the formation of barest conditionals.

KEYWORDS

Barest conditional Identical pairing Unselective binding Prosodic fusion Syntax-prosody interface Modal licensing

1. INTRODUCTION

The barest conditionals discussed in this paper are drawn from the first two types of the Mandarin bare conditionals identified by Tsai (2025).

Type I: *Gāi*-conditionals

- (1) *Gāi chī (fàn) chī (fàn), gāi hē (jiǔ) hē (jiǔ).*
 GAI eat.meal eat.meal GAI drink.wine drink.wine
 ‘Eat if it’s time to eat, drink if it’s time to drink.’
- (2) *Gāi gōngzuò gōngzuò, gāi xiūxi xiūxi.*
 GAI work work GAI rest rest
 ‘Work if it’s time to work, rest if it’s time to rest.’

Type II: *Wh*-conditionals

- (3) *Ài shéi shéi.*
 AI who who
 ‘Let whoever it may be.’
- (4) *Ài zǎbàn zǎbàn.*
 AI how.do. how.do
 ‘Handle it however you like.’

As demonstrated above, the first type of barest conditionals is introduced by the deontic modal verb *gāi* ‘ought to.’ The second type is introduced by *ài* ‘to love/want’. The second type is essentially a special kind of “donkey sentences” as discussed by Cheng & Huang (1996, 2020), where an identical pair of *wh*-expressions is employed, as illustrated below (Cheng & Huang 1996: 127):

- (5) *Shéi xiān lái, shéi xiān chī.*
 who first come who first eat
 ‘Whoever comes first gets to eat first.’

Here we dub (1-4) “barest conditionals”, distinguishing them from the bare one in (5) in view of their even more reduced profiles and the absence of prosodic boundaries between the antecedent and consequent clauses. Specifically, these barest conditionals possess the following formal characteristics:

First, barest conditionals dispense with subordinating conjunctions that typically signal the logical relationship between the antecedent and the

consequent clause, e.g., *rúguǒ* ‘if’, *zhǐyào* ‘if only’ for the former, and *jiù* ‘then’ for the latter.¹

From a typological perspective, these conditionals fall under the category of asyndetic conditionals. Secondly, these constructions are structurally streamlined by keeping only the absolutely necessary components in both the antecedent and consequent clauses. What remains are typically either a pair of bare verbs or VPs (in the first type) or a pair of *wh*-expressions (in the second type), subject to the so-called “identical pairing” (see Cheng & Huang 2020: 169, on “equal siblings”; Hole 2004: 24, on “twin variables”).

Thirdly, from a prosodic point of view, barest conditionals display a strong preference for prosodic symmetry and balance between the antecedent and consequent clauses (see Yang & Tsai 2022). It is also noteworthy that the prosodic boundary between the two clauses is entirely absent. As illustrated in (6), no pause or pause particle like *a* may be inserted between the antecedent and the consequent.

- (6) **Ài shéi ... shéi.* / **Ài shéi a shéi.*
 AI who PAUS who AI who PRT who
 Intended: ‘Let whoever (it is) be.’

Tsai (2025) distinguishes between the two types of barest conditionals primarily based on their surface structural differences and further proposes that the two types are licensed by slightly different syntactic mechanisms. First, despite the existence of identical pairing, the paired elements are verbal expressions in the first and *wh*-expressions in the second. As such, the second type is also dubbed *wh*-conditionals, which closely resemble the bare conditionals in the sense of Cheng & Huang (1996). Secondly, they also differ in their sentence-initial licensers. The first type is introduced by the deontic modal particle *gāi* at the sentence-initial position, which serves as a necessity operator. It is important to note that the barest conditionals in (1) and (2) do not contain any lexical variables such as *wh*-expressions or indefinite pronouns.

In contrast, the second type begins with the verb *ài* ‘to love/want’, whose syntactic and semantic functions will be analyzed in detail in the following sections. Based on the identical pairing of *wh*-elements in

question, Tsai argues that the pair of *wh*-variables are bound by an implicit necessity operator along the line of Cheng & Huang (1996), resulting in the barest *wh*-conditionals in (3) and (4).

In sum, while both types of barest conditionals are licensed via unselective binding by a necessity operator, they differ in whether to pronounce their licensing operators at PF: The first type is licensed by an explicit necessity operator, which is realized by the deontic modal *gāi*, whereas the second is licensed by a silent one. In particular, Tsai does not identify sentential-initial *àì* as a conditional operator.

Based on Cheng & Huang (1996, 2020) and Tsai (2023, 2025), this article aims to address several mysteries surrounding the licensing mechanism of barest conditionals: Firstly, following Tsai's analysis, both types of conditionals are licensed by some form of necessity operator through unselective binding. However, it is unclear how the classic analysis may apply to both types of barest conditionals, especially considering their different pairing patterns. Also, it is equally unclear why we should treat the sentence-initial elements of these two types differently, given that structurally they occupy the same position and are equally indispensable. We will illustrate the licensing mechanism of barest conditionals in section 2. Secondly, what motivates the extreme structural reduction found in barest conditionals? We contend that this question leads to another crucial dimension of these constructions, i.e., prosody. Specifically, we plan to explain why the antecedent and consequent in barest conditionals are so closely bound in prosodic terms, such that no pause or pause particle is permitted between them. These prosodic constraints will be discussed in section 3.

2. THE SYNTACTIC LICENSING MECHANISM OF BAREST CONDITIONALS

2.1 CONDITIONALS LICENSED BY AN EXPLICIT NECESSITY OPERATOR

A close examination of Cheng & Huang's (1996, 2020) unselective binding analysis reveals that the motivation for introducing an implicit necessity operator in bare conditionals lies in the nature of Chinese *wh*'s-in-situ. Namely, they behave like polarity items, which lack inherent quantificational force. As such, they function as variables that require

binding and interpretive support from an external quantificational operator, such as a quantificational adverb, a modal, or an implicit necessity operator. Cheng & Huang propose that this (implicit) necessity operator functions analogously to a universal quantifier. It unselectively binds variables with matching referential indices across the antecedent and consequent clauses. Once the *wh*-variable is bound, the resulting interpretation is one of universal quantification, as illustrated in example (7) (Cheng & Huang 2020: 169).²

- (7) a. [_{ANT} ... *wh* ...], [_{CONQ} ... *wh* ...]
 b. [_{NECX} [_{ANT} ... *wh_x* ...], [_{CONQ} ... *wh_x* ...]]
 c. [_{∀X} [... *x* ...] → [... *x* ...]]

Based upon Tsai (2023, 2025), we argue that the two types of barest conditionals discussed above share the exact licensing mechanism: Both sentence-initial modals serve as a lexical necessity operator that unselectively bind the pair of variables in the antecedent and consequent clauses, thereby enabling the barest conditional construal in question. A natural question arises: What exactly is this explicit operator?

As Tsai (2025) observes, in the first type of barest conditional, the sentence-initial modal *gāi* serves this function. Given that *gāi* is an obligatory modal, it is reasonable to treat it as a necessity operator for the conditional construal in question. Furthermore, Tsai proposes that *gāi*-conditionals are built upon a pair of situation variable across the antecedent and consequent clauses.

- (8) *gai_x* [_{pro_i} eat (meal) in situation <*x*>] [_{pro_i} eat (meal) in situation <*x*>]
gai_y [_{pro_i} drink (alcohol) in situation <*y*>] [_{pro_i} drink (alcohol) in situation <*y*>]

This way, the licensing mechanism of *gāi*-conditionals is consistent with Cheng & Huang's analysis. The only crucial difference lies in the type of variable being bound: In a *gāi*-conditional such as (1), the necessity operator binds a pair of "situation variables" rather than *wh*-variables.³ Notably, Tsai analysis of situation variables is akin to Chierchia's (1992:

147) notion of “occasion variables”. According to Chierchia, when a conditional lacks an indefinite noun phrase and the event expressed by the verb becomes the discourse topic, the occasion variable associated with that event is activated and becomes available for quantificational binding. In the case of *gāi*-conditionals, since the only available material consists of verbs under the information focus, the occasion/situation variables become activated in syntax for operator binding. This observation can be supported by the fact that, as noted by one reviewer, in *gāi*-conditionals with VP-pairing such as (1) and (2), it is possible to append the situation-marking expression *de shíhòu* ‘when’ to the VP (e.g., *Gāi chī (fàn) de shíhòu, chī (fàn), gāi hē (jiǔ) de shíhòu, hē (jiǔ)*), whereas this is not possible in *wh*-type conditionals. Although the addition of *de shíhòu* changes the structure of (1) and (2), so they are no longer strict barest conditionals, and a pause can indeed be inserted between the two VPs in such cases, the contrast still provides semantic evidence that a situation variable is involved in VP-type barest conditionals, where a *wh*-variable is not available. We will revisit this point in 2.3.

How, then, does the second type of barest conditional differ from the first? The second type of barest conditional does not seem to contain an overt modal. However, since *wh*-expressions appear in both the antecedent and consequent clauses, these conditionals can be analyzed in the same way as bare conditionals à la Cheng & Huang, namely, with identical *wh*-variables bound unselectively by an implicit necessity operator. Tsai (2025) treats the sentence-initial *ài* ‘to love’ as a verb scoping only over the *wh*-variable in the antecedent clause, as illustrated in the representation below:

- (9) $NEC_{x,y}$ [pro_i love who(y) in situation $\langle x \rangle$] [pro_i (look for) who(y) in situation $\langle x \rangle$]

Building on Tsai’s analysis, we propose that the licensing mechanism of the second type of barest conditional is identical to that of the first type. Specifically, it is claimed that the operator licensing conditional construals is actually a lexical modal, i.e., the sentence-initial *ài*. In what follows, we offer a brief analysis of *ài* in this modal usage.

The word *ài* exhibits different semantic and functional properties depending on the context. In the current construction, *ài* should not be interpreted in its canonical sense as a psychological verb meaning ‘to love’, but rather as a bouletic modal, expressing the subject’s willingness, desire, or preference. Consider the following examples:

- (10) *Nǐ ài qù nǎr, jiù qù nǎr.*
 you AI go where then go where
 ‘You may go wherever you would like.’
- (11) *Nǐ ài zhǎo shéi, jiù qù zhǎo shéi.*
 you AI find who then go find who
 ‘Go find whoever you feel like finding.’

Clearly, here *ài* means something akin to *want* or *wish*, and thus should be analyzed as a bouletic modal.⁴ In appropriate discourse and prosodic contexts, these conditionals are interpreted very much like their barest counterparts, such as *ài nǎr nǎr* ‘wherever you want’, and *ài shéi shéi* ‘whoever you want’.

Since the sentence-initial element is analyzed as a modal, can it function as an overt licenser of the second type of barest conditionals? The key issue is whether bouletic modals, modals expressing desire or volition, can perform the same role as a necessity operator. As discussed earlier, necessity operators function akin to universal quantifiers, yielding universal readings through unselective binding. There is evidence suggesting that bouletic modals may indeed give rise to universal quantification when their modal force is interpreted over a space of possible worlds, thus producing the necessity interpretation required (Wimmer 2022; see also Heim 1992; Rubinstein 2017). Rubinstein (2017), following Heim (1992), argues that *want*-like attitude verbs should be treated as universal quantifiers over possible worlds, just like necessity modals. Along similar lines, Wimmer (2022: 419) provides a unified analysis of the Mandarin modal verb *yào* (要) ‘to want; to need’. *Yào* displays a well-known duality, expressing both desire/intention and obligation/necessity. Wimmer suggests that these distinct readings arise from differences in modal flavor rather than modal force: the core meaning remains universal quantification over possible worlds. More concretely,

yào always takes two propositions p and q , such that p restricts quantification over q , and all p -worlds are q -worlds.

Further support for the “force operator” view comes from the observation, as suggested by one of our reviewers, that *ài*-conditionals often convey a complaining nuance, in addition to expressing universal force. This is comparable to the “whining force” operator found in *V shénme (NP)* constructions (e.g., *mányuàn shénme?! ‘Complaining what?!’*) (Tsai 2011, 2021; Yang & Tsai 2019). The phenomenon may be understood as a case of “force shift,” akin to the shift from interrogative to whining force in *wh*-expressions. By analogy, *ài* must function as a force operator to undergo such interpretive shifts. Under our analysis, *ài* is best characterized as a strong modal force operator with universal quantificational force, parallel to a necessity operator. This necessity-based nature forms the basis for further shifts in interpretation, including the transition from a necessity meaning to a complaining or whining reading. Such shifts also align with possible grammaticalization pathways of modal operators (see 2.4 for further discussion).

Following the unified approach of bouletic modal and the necessity modal, we can say that, in terms of quantificational force, either a deontic modal (such as *gāi*) or a bouletic modal (such as *ài*) may be classified as a necessity operator. That is, while the two types of modals differ in their modal base or modal flavor (i.e., the source of modality, one based on rules while the other based on desires), they are alike in expressing a strong modal force, i.e., necessity. Semantically, both may be analyzed as a universal quantifier scoping over possible worlds: Although the barest conditionals are licensed by distinct modals, they eventually converge: Both types of modals serve to bind and license variables in the structure, imparting a necessity interpretation to the conditional as a whole. (For further discussion of the universal semantics of these conditionals, see below.)

Now we have established that both types of modals carry universal quantificational force, hence are qualified as the licensers of barest conditionals. This means that they implement the licensing mechanism in the same manner: Both modals should be able to bind either the paired situation variables (introduced by verbs) or the paired *wh*-variables. Our prediction is borne out with the following data:

- (12) a. *Tīngshuō Zhāng Sān yào qù Měiguó le.*
 hear-say Zhang San will go USA ASP
 ‘I heard Zhang San is going to the U.S.’
 b1. *Ài qù qù, wǒ cái bù guǎn ne.*
 AI go go I EMPH not care PRT
 ‘Let him go if he wants, I really don’t care.’
 b2. *Ái shéi (qù) shéi (qù), wǒ cái bù guǎn ne.*
 AI who (go) who (go) I EMPH not care PRT
 ‘Whoever wants to go can go—I really don’t care.’
- (13) a. *Dé-le zhè-ge bìng, xūyào jìkǒu ma?*
 get.ASP this.CL illness need avoid.food Q?
 ‘If I get this illness, do I need to avoid certain foods?’
 b. *Nǐ gāi chī chī, gāi hē hē,*
 you GAI eat eat drink drink drink
bù yòng jìkǒu.
 not need avoid.food
 ‘Just eat and drink as usual, no dietary restrictions needed.’
- (14) a. *Shéi lái wánchéng zhè-ge rènwù?*
 Who come complete this. CL task
 ‘Who comes to complete this task?’
 b. *Gāi shéi shéi.*
 GAI who who
 ‘Whoever should (do it), does (it).’

The sentences above suggest that both types of conditionals permit an identical pairing of either verbs or *wh*-elements. We argue that the determining factor for the exact realization of identical pairing is which variable functions as the sentence’s information focus, or what Chierchia (1992:147) refers to as the “discourse topic”. Specifically, when the speaker intends to emphasize the necessity of the event itself, the event becomes the information focus. In this case, the situation variable is activated, forming a verb-based identical pairing, as seen in (12b1) and (13b). By contrast, when the focus is placed on the *wh*-word, the *wh*-variable is activated, resulting in the *wh*-based identical pairing in (12b2) and (14b). We will return to this issue in 2.3.

Along the line laid out above, the syntax and semantics of these two types of barest conditionals can be schematized as follows:

***Gāi*-type barest conditionals:**

- (15) Verb-based identical pairing, e.g., *Gāi chīfàn chīfàn*. ('If (you) ever need to eat, then eat.')

DEO [ANTE...VP...] [CONS...VP...]
[NEC_x [ANTE...SIT_x...VP...] [CONS...SIT_x...VP...]]

Its universal quantificational semantics can be expressed as:

For all (minimal) situations x under consideration, if in any situation x Pro is obligated to eat, then it is necessary for Pro to eat in situation x.

- (16) *Wh*-based identity pairing, e.g., *Gāi (chī) shénme (chī) shénme*. ('(You) should eat whatever (you) need to eat.')

DEO [ANTE...wh...] [CONS...wh...]
[NEC_x [ANTE...wh_x...(VP)...] [CONS...wh_x...(VP)...]]

Its universal quantificational semantics can be expressed as:

For all x such that x is something Pro is obligated to eat, it is necessary for Pro to eat x.

***Ài*-type barest conditionals**

- (17) Verb-based identical pairing, e.g., *Ài qù qù*. ('If (you) ever want to go, go.')

BOUL [ANTE...VP...] [CONS...VP...]
[NEC_x [ANTE...SIT_x...VP...] [CONS...SIT_x...VP...]]

Its universal quantificational semantics can be expressed as:

For all (minimal) situations x under consideration, if in any situation x Pro wants to go somewhere, then it is necessary for Pro to go there in situation x.

- (18) *Wh*-based identity pairing, e.g., e.g., *Ài (qù) nǎlǐ (qù) nǎlǐ*. ('Wherever (you) want to go, go.')

BOUL [ANTE...wh...] [CONS...wh...]
[NEC_x [ANTE...wh_x...(VP)...] [CONS...wh_x...(VP)...]]

Its universal quantificational semantics can be expressed as:

For all x such that x is a place Pro wants to go, it is necessary for Pro to go to x.

Based on the above analysis, we may summarize the licensing mechanism of barest conditionals as follows: Barest conditionals sport two types of identical pairing: When the event is in the information focus, the clause-initial modal functions as a necessity operator binding a pair of identical situation variables in both the antecedent and the consequent, thereby yielding the conditional construal of universal quantification. In this case, we get an identical pairing of two verbal expressions. When the question is focused, the clause-initial modal binds a pair of *wh*-variables, again producing the conditional interpretation. We thus get an identical

pairing of two *wh*-expressions instead. Note that different modals exhibit distinct lexical meanings: *gāi* conveys obligation, while *ài* expresses desire. They appear only in the antecedent clause. On the other hand, as an unselective binder, these modals always scope over both antecedent and consequent clauses, resulting in the global necessity interpretation in question (cf. Tsai 2023).

Along this line, we propose that the two types of barest conditionals should receive a unified treatment: Namely, they are structurally reduced versions of bare conditionals, where the licensing mechanism aligns closely with the one proposed by Cheng & Huang. If our analysis is on the right track, the diagnostic tests that Cheng & Huang employ to support the configuration of “necessity operator + identical variables” should apply to barest conditionals as well. In what follows, we present empirical evidence using their diagnostics.

2.2 EVIDENCE THAT BAREST CONDITIONALS EMPLOY UNSELECTIVE BINDING

Cheng & Huang (1996, 2020) argue that anaphoric elements in conditional sentences can be interpreted via two distinct strategies. The first is unselective binding by a necessity operator, as in the case of bare conditionals discussed earlier. The second is the E(vans)-type strategy, in which the interpretation of the anaphoric element proceeds descriptively and is inferred from contextual or pragmatic cues (cf. Evans 1980: 340). For instance, in canonical *if*-conditionals, pronouns, null pronouns, or definite noun phrases in the consequent are typically interpreted through this E-type mechanism.

The following syntactic diagnostics further support the claim that barest conditionals share the same licensing mechanism as the bare conditionals analyzed by Cheng & Huang.

Diagnostic 1: Incompatibility with the Existential Marker *yǒu*

Cheng & Huang (1996: 141–142) observe a contrast between canonical *if*-conditionals and bare conditionals regarding the admissibility of the existential marker *yǒu* ‘to have’ in the antecedent clause. Specifically, in bare conditionals like (19), *yǒu* is not allowed, whereas it is permissible in *if*-conditionals, as shown in (20):

- (19)* *Yǒu shéi xiān lái, shéi xiān chī.*
 have who first come who first eat
 Intended meaning: ‘Whoever comes first, eats first.’
- (20) *Rúguǒ yǒu shéi xiān lái, shéi jiù xiān chī.*
 if have who first come who then first eat
 ‘If someone comes first, that person will eat first.’

This contrast indicates that the *wh*-elements in the antecedents of the two types of conditionals are interpreted differently. In canonical *if*-conditionals (e.g., (20)), the *wh*-element in the antecedent can be interpreted existentially and is licensed and bound within the conditional clause itself by the conditional marker *rúguǒ* ‘if’. This allows the antecedent to constitute a complete tripartite structure internally (cf. Kratzer 1989), without requiring an external necessity operator to bind variables across clauses. In contrast, bare conditionals rely on unselective binding by a necessity operator to license the *wh*-variable.

If the barest conditionals under discussion follow the same unselective binding strategy proposed by Cheng & Huang for bare conditionals, then they should similarly disallow the existential marker *yǒu*. The following test confirms this prediction, which demonstrates that in barest conditionals, both *wh*-variables and situation variables cannot be interpreted existentially.

- (21) a. *Ài (*yǒu) zhǎo (shéi) zhǎo shéi.*
 AI have find who find who
 b. *Ài (*yǒu) shéi shéi.*
 AI have who who
- (22) a. *Gāi (*yǒu) (shéi) chī (shéi) chī.*
 GAI have who eat who eat
 b. *Gāi (*yǒu) shéi shéi.*
 GAI have who who

Diagnostic 2: No *wh*/pronoun alternations

As noted in Footnote 1, *wh*-elements in the consequent clause of canonical *if*-conditionals can be replaced by other nominal expressions, such as pronouns or definite noun phrases. This difference arises from the distinct variable-binding strategies discussed earlier: In *if*-conditionals, *wh*-variables in the antecedent are not bound by a necessity operator and

therefore do not require an identical *wh*-form to appear in the consequent clause. In contrast, bare conditionals and the barest conditionals under discussion do not permit such alternations. This is illustrated by (23)–(25):

- (23) *If*-conditional:
Shéi yào zhè pò-chē,
 who want this broken.car
wǒ jiù ràng gěi shéi/tā/nà-ge-rén.
 I then let give who/he/that.person
 ‘Whoever wants this junk car, I’ll give it to them/him/that person.’
- (24) Bare conditional:
*Shéi xiān lái, shéi/*tā/*nà-ge-rén xiān chī.*
 who first come who/he/that.person first eat
 ‘Whoever comes first eats first.’
- (25) Barest conditional:
*Ài shéi shéi/*tā/*nà-ge-rén.*
 AI who who/he/that.person
 ‘Let whoever (it is) be.’

These results further support the claim that *wh*-variables in barest conditionals cannot be analyzed as existentially quantified expressions nor as E-type pronouns. Instead, they must be treated as variables that require unselective binding by a necessity operator.

Diagnostic 3: No different licensing strategies in one sentence

Previous research (Cheng & Huang 1996, 2020; Lin 1996; Pan & Jiang 2015) has observed that in *if*-conditionals or conditionals with *jiù* ‘then’, it is possible for two types of anaphoric elements, each licensed by a distinct strategy, to co-occur. Consider the example in (26):

- (26) (*Rúguǒ*) *shéi xiǎng chī shénme, tā jiù chī shénme.*
 if who want eat what he then eat what
 ‘(If) whoever wants to eat something, he eats that thing.’

According to Cheng & Huang’s analysis, this sentence exhibits two different interpretive strategies: *Shéi* ‘who’ in the subject position of the antecedent is interpreted existentially, licensed by the conditional marker *rúguǒ* ‘if’, and its counterpart *tā* ‘(s)he’ in the consequent is interpreted as an E-type pronoun. Meanwhile, the object-position *wh*-word *shénme* is

bound unselectively by a necessity operator and must be paired by an identical *wh*-form in the consequent clause.

In barest conditionals, although multiple pairings may appear simultaneously in different syntactic positions, as seen in (27), it is not possible for the sentence to contain pairings licensed by different strategies, as seen in (28), where the second *shéi* ‘who’ is replaced by *tā* ‘he’, and (29) where the second *shénme* ‘what’ is replaced by *tā* ‘it’.

- (27) *Gāi shéi (chī) shénme shéi (chī) shénme.*
 GAI who eat what who eat what
 ‘Whoever should eat what, eat what (Each person eats what they are supposed to eat).’
- (28)* *Gāi shéi (chī) shénme tā (chī) shénme.*
 GAI who eat what he eat what
- (29)* *Gāi shéi (chī) shénme shéi (chī) tā.*
 GAI who eat what who eat it

These data indicate that, in barest conditionals, only one licensing mechanism is available, namely, the unselective binding strategy enforced by the necessity operator. The use of mixed strategies, as permitted in *if*-conditionals, is not possible in this construction.

2.3 AN ACCOUNT TO TWO VARIATIONS OF IDENTICAL PAIRING

Why, then, must identical pairings appear in both bare conditionals and barest conditionals? Cheng & Huang (1996) account for this requirement in bare conditionals by appealing to the Prohibition Against Vacuous Quantification (PAVQ), initially proposed by Kratzer (1989: 155):

- (30) For every quantificational operator *Q*, there must exist a variable *x* such that *Q* binds *x* in both its restrictive clause and its nuclear scope.

According to this principle, to ensure semantic well-formedness and prevent the operator from becoming vacuous, the necessity operator in the conditional must unselectively bind at least one variable that is present in both the antecedent (restrictive clause) and the consequent (nuclear scope). To explain why identical *wh*-elements, rather than pronouns or null categories, must appear in both clauses, Cheng & Huang (1996: 139) propose an enhanced version of the Parallelism Constraint on Operator

Binding (PCOB), called the Revised PCOB, which is based on Safir's (1985) original formulation.

- (31) In a tripartite structure of quantification $Q [A] [B]$, $[X_1, X_2, \dots, X_n]$ (where $n \geq 1$) are variables present in $[A]$. For every variable in A , there must be an identical variable in $[B]$.

This strengthened version not only requires the variables bound by the same operator to exhibit syntactic and semantic parallelism, but it also imposes a morphological identity condition on their realization. That is, the variables must be morphologically identical in form. Cheng & Huang (2020: 182) illustrate this with the following example:

- (32) *Shéi xiān lái shéi / *shénme-rén xiān chī.*
 who first come who / *what.person first eat
 'Whoever comes first, eats first.'

Although *shéi* 'who' and *shénme-rén* ('what person') are both interrogative expressions and may be interpreted as coreferential at the syntactic and semantic levels, the sentence becomes unacceptable when morphological identity is violated. This follows directly from the revised PCOB, which stipulates that only form-identical pairings are licit.

As we have argued earlier, barest conditionals adopt the same strategy of unselective binding by a necessity operator. Consequently, they too are subject to the revised PCOB and must satisfy the morphological identity requirement. This prediction is borne out in the following examples:

- (33) *Gāi chī shénme chī shénme / *nǎge dōngxi.*
 GAI eat what eat what / which thing
 '(You) should eat whatever (you're supposed to eat).'
- (34) *Ài zhǎo shéi zhǎo shéi / *shénme rén.*
 AI find who find who / what person
 '(You) can go to whoever you want.'

On this basis, we can say that the syntactic motivation for the obligatory pairing in both bare and barest conditionals can be traced to two interrelated constraints: at LF (Logical Form), the necessity operator's

unselective binding of variables must satisfy the Prohibition Against Vacuous Quantification (PAVQ), while at PF (Phonological Form), the binding must obey the strict version Parallelism Constraint on Operator Binding (PCOB). Since these constraints do not discriminate between types of variables, they can naturally be extended from *wh*-variables to situation variables. This generalization predicts that in barest conditionals, verbal phrases expressing situation variables should likewise be subject to identical pairing in their forms. This prediction is indeed borne out: In barest conditionals where the event itself is the focus of information, the event must be expressed not only with semantic identity but also with morphological identity. For example, in (35) and (36), although the verb phrases in the antecedent and consequent may refer to the same kind of event, their distinct lexical forms violate the revised PCOB and render the sentences unacceptable:

- (35) **Gāi xiūxi fàngsōng*.
 GAI rest rest
 Intended: ‘(You) should rest if it’s time to rest.’
- (36) **Ài lǚyóu lǚxíng*.
 AI travel travel
 Intended: ‘Travel whenever you want.’

However, extending the unselective binding and the revised PCOB to situation variables introduces a theoretical challenge, as noted in footnote (2): If situation variables are indeed bound in the same manner as *wh*-variables in barest conditionals, for instance in (5), then why do we not observe identical verbal phrase pairing in such constructions? In other words, if all bound variables must be overtly realized in morphologically identical form, as the revised PCOB stipulates, then a well-formed version of a bare conditional such as (5) would have to appear as *Shéi xiān lái, shéi xiān lái*. This, however, clearly does not align with empirical facts of the language.

Chierchia’s (1992) discussion offers valuable insight: He proposes that only those variables that constitute the discourse focus are obligatorily bound by quantificational operators. For instance, when the event itself is the discourse focus, its associated implicit “occasion operator” is bound by the necessity operator. In such cases, barest conditionals surface with

identical VP pairings. Conversely, when the *wh*-elements are in focus, it is the *wh*-variables that are realized in the PF, yielding identical *wh*-pairings. While we maintain that necessity operators, in principle, unselectively bind all variables within their scope (whether situation or *wh*-variables), not all variables must be morphologically realized as identical pairs at once. Building on Chierchia's insight, we suggest that the variables that surface as identical pairs are precisely those that serve as information foci, in order to satisfy the strengthened PCOB, which requires morphological identity at PF.

The relationship between information focus and the selection of VP pairing and *wh*-pairing can be shown in (37), where OP_{Nec} indicates the necessity operator carrying modal force $\forall$ across worlds, and F indicates the [+F], the focus-marking feature.

- (37) **Case 1: Event focus $\rightarrow$ VP pairing**
 $OP_{Nec} [F SIT_x] \rightarrow \langle VP_1, VP_2 \rangle, VP_1 = VP_2$

Case 2: *Wh*-focus $\rightarrow$ *wh*-pairing
 $OP_{Nec} [F WH_x] \rightarrow \langle WH_1, WH_2 \rangle, WH_1 = WH_2$

This explains the two distinct types of overt identity pairings observed in barest conditionals. Furthermore, in cases where multiple variables are simultaneously in focus, multiple identical pairings may surface. For example, sentence (38) displays two *wh*-pairs, both of which are information foci: Suppose you hear from someone that on the big shopping day, everyone buys different things: A bought this, B bought that. You might respond:

- (38) *Ài shéi mǎi shá shéi mǎi shá,*
 AI who buy what who buy what
fǎnzhèng wǒ bù mǎi.
 anyway I not buy
 'Let whoever wants to buy whatever buy it. I'm not buying it anyway.'

Additionally, in sentences containing both *wh*-variables and situation variables, *wh*-elements are more likely to be the information focus. Therefore, the morphological identity requirement (as per PCOB) applies

more strictly to *wh*-variables, while the identical pairing of situation variables (e.g., verb phrases) may be relaxed, especially when there is also *wh*-pairing in the sentence. Consider the following examples:

- (39) *Gāi zǎ-(bàn) zǎ-bàn.*
 GAI how.do how.do
 ‘Just deal with it however it should be dealt with.’

In these cases, as long as the *wh*-variable *zǎ* ‘how’ is identically paired, the verb part *bàn* ‘to do’ can be elided in the antecedent, reflecting the looser constraint on non-focus elements.

Our proposal also accounts for an interesting phenomenon in Northeastern Mandarin observed by Tsai (2025): The *wh*-item *zǎ-dì* appears to deviate from strict identity pairing. For example:

- (40) *Ài zǎ(dì) zǎdì.*
 AI how.do how.do
 ‘Let it be however it wants (=Whatever)!’

Here, it is crucial to recognize that *zǎ-dì* in Northeastern Mandarin is not equivalent to the *zǎ* ‘how’. Rather, it functions more like a compound of a *wh*-word and a light verb, roughly glossed as ‘how-be’ or ‘how-do’, akin to Standard Mandarin *zěnme-nòng* or *zěnme-zhāo*. Thus, in example (39), as long as the true *wh*-variable *zǎ* is identically paired, which satisfies the strengthened PCOB, the light verb component *dì* may be exempt from strict morphological identity such as the omission of its first instance. However, as Tsai notes, although forms like (39) are acceptable, reversing the order (e.g., **Ài zǎ-dì zǎ*) results in unacceptability, likely due to prosodic constraints.⁵

2.4 CONSTRAINTS ON MODAL VERBS LICENSING BAREST CONDITIONALS

As noted in the previous discussion, the modal verbs *gāi* ‘ought to’ and *ài* ‘want’ function as necessary operators that license barest conditional constructions. This observation naturally leads to the following question: Which modal verbs can license such constructions?

An examination of the two sets of barest conditional sentences in (41) and (42) reveals that the modal verbs that can appear in these constructions tend to occupy relatively low syntactic positions. These include deontic modals, such as *gāi* ‘should’, and dynamic modals, such as *ài* ‘like’, *xiǎng* ‘want’, *yào* ‘need’, *kěn* ‘be willing’, *néng* ‘be able’, and so forth.⁶ In contrast, epistemic modals that occur in higher syntactic positions, such as *yīnggāi* ‘it should be the case that’, fail to license barest conditionals, as evidenced by the unacceptability of the examples in (43).

- (41) *Ài/gāi/xiǎng/yào/kěn/gǎn/néng chī chī.*
 AI/GAI/XIANG/YAO/KEN/GAN/NENG eat eat
 ‘Eat whenever you feel like/should/want/need/are willing/are able (to) eat(ing).’
- (42) *Ài/gāi/xiǎng/yào/kěn/gǎn/néng (zhǎo)-shéi (zhǎo)-shéi.*
 AI/GAI/XIANG/YAO/KEN/GAN/NENG find.who find.who
 ‘Find whoever you feel like/should/want/need/are willing/are able (to) find(ing).’
- (43)* *Yīnggāi^{epistemic} (huì) chī chī / shéi shéi.*
 it should be the case eat eat who who

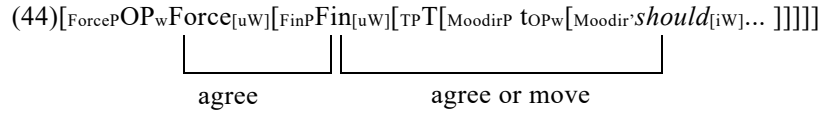
One interesting observation is that most of the licensing modals in barest conditionals are monosyllabic.⁷ Why are monosyllabic modal words more likely to appear in barest conditionals? Given the highly colloquial nature of these constructions, a plausible explanation is that modal words entering such structures must also exhibit a high degree of colloquialization to ensure stylistic coherence. In many cases, monosyllabic expressions in Chinese tend to be more colloquial and informal than their disyllabic counterparts with similar meanings (cf. Feng & Shi 2023: 367). However, there also exist monosyllabic modal verbs that are entirely excluded from this structure, such as *yīng* (应) ‘should’, *bì* (必) ‘must’, and *xū* (须) ‘need’. We argue that this exclusion stems from the fact that these modal verbs are typically restricted to written or formal registers. These observations reflect a fascinating interaction among prosody, register, and syntax, which is exhibited in this construction. Now let’s turn back to our syntactic observations.

If we compare the modal verbs that license barest conditional constructions with the three-tiered modal hierarchy proposed by Tsai (2010, 2015), a clear pattern emerges: The modals that can license barest

conditionals tend to originate in the inflectional layer, or even lower in the lexical layer, while modal verbs located in the complementizer layer are generally excluded from this licensing capacity. We argue that this phenomenon reflects the movement pathway from modals to conditional operators: Low-position modals, or root modals, undergo syntactic movement upward into a higher structural position (cf. Roberts & Roussou 2003). Specifically, these modals, initially located in projections such as MP^{Dyn} and MP^{Deo} , move upward toward the syntactic position associated with the conditional operator, commonly identified as the $Mood_{irrealis}$ in Cinque's (1999) universal hierarchy (see also Haegeman 2010). Once occupying this position, the modal acquires the capacity to license conditional constructions. This analysis accounts for why only low-position modals can serve as licensing operators in conditional constructions: According to Cinque (1999: 106), the $Mood_{irrealis}$, which hosts the conditional operator, is located within the TP/IP domain, above the layers associated with root modals such as MP^{Dyn} and MP^{Deo} , but below epistemic modals, which reside higher in the hierarchy. While lower modals can move upward to become conditional operators, higher epistemic modals do not typically descend to take on this role. Interestingly, Haegeman (2010: 606) makes a parallel observation in her study of English and other languages: Conditional clauses are generally incompatible with elements that originate above $Mood_{irrealis}$, such as epistemic modals and evidential mood particles.

Furthermore, there is reason to believe that once certain modal verbs are reanalyzed as conditional operators, they may undergo further upward movement, allowing them to interact with higher speaker-oriented projections, thereby giving rise to subjective or subjunctive interpretations in addition to the conditional reading. Breitbarth, Delva, & Leuschner (2016) examined a set of asyndetic conditionals across English, German, and Dutch, in which modal verbs function as conditional operators. Building on Haegeman (2010: 608–609), they propose a syntactic pathway whereby modals originate in their base-generated positions but subsequently move into the $Mood_{irrealis}$ projection to serve as conditional operators. From there, these modals can further interact with higher functional projections such as FinP or even ForceP at the left periphery, either through Agree or Move, in order to receive “world feature” valuation.

This proposed movement and feature interaction are schematized in the following syntactic hierarchy (Breitbarth, Delva, & Leuschner 2016: 306):



Our data show that among all modal verbs that can license barest Conditionals in Mandarin, the two most productive ones are *gāi* and *ài*, which have been the main focus of our discussion thus far. These two modals can not only combine with low-position verbal phrases but also with higher-position interrogative elements. In contrast, other modals lack this syntactic versatility, as illustrated in (45):

- (45) *Ài* / *Gāi* / **xiǎng* / **yào* / **kěn* / **gǎn* / **néng* *shéi* *shéi*.
 AI/GAI/XIANG/YAO/KEN/GAN/NENG who who

This pattern indicates that *ài* and *gāi* may have moved into the CP layers, allowing them to directly associate with interrogative variables in both conditional and clausal structures. Interestingly, expressions like *Ài shéi shéi* and *Gāi shéi shéi* also carry pragmatic meanings beyond their basic universal quantificational force. They often convey a speaker’s dismissive or indifferent attitude, i.e., expressing a “whining force” as discussed in 2.1 (see Tsai 2025), which we interpret as the pragmatic outcome of their speaker-oriented usage as high conditional operators, possibly resulting from grammaticalization and the interaction between these conditional operators and the ForceP in the CP layer, as shown in (44).⁸

3. PROSODIC FUSION AS A DRIVING FORCE OF STRUCTURAL REDUCTION

As mentioned in section 1, in barest conditionals, the antecedent and the consequent must be tightly connected without any pause or prosodic break in between. This section explores the prosodic constraints that shape barest conditionals, and we will argue that the prosodic fusion of the bipartite conditionals is the driving force of the structural reduction in

barest conditionals.

In regular conditionals marked with *rúguǒ...jiù...* ‘if...then...’, a natural prosodic break or pause often exists between the antecedent and the consequent. In contrast, in bare conditionals or barest conditionals, the two clauses are highly compact and do not allow for pauses or interjectional particles such as *a* in between, giving the impression that the two clauses have fused into a single prosodic unit. Cheng & Huang (2020: 179) describe this phenomenon as forming a “compact unit.” We propose that this prosodic fusion between antecedent and consequent clauses leads to the disappearance of prosodic boundaries, which in turn motivates syntactic simplification: The two clauses effectively become one, further facilitating the emergence of barest conditionals in spoken Mandarin.

There are many examples in Chinese where changes in prosodic structure drive changes in syntactic structure. Feng (2017) argues that prosodic restructuring is a fundamental driving force in the syntactic evolution of Chinese. He points out that within a single sentential intonation phrase (SIP) in Chinese, only one nuclear stress is permitted; consequently, there can only be one predicative verb per sentential intonational unit.⁹ Feng refers to this as the “Constraint on Sentential Intonation in Chinese”. This one-to-one correspondence between intonational phrase and predicate (一句一调, 一调一动/*yíjù yídiào, yídiào yídòng*) has a significant impact on Chinese syntax, one result of which is the emergence of what are known as “serial verb constructions” (连动式/*liándòng shì*) and “flowing sentences” (流水句/*liúshuǐ jù*). He cites Chao (1968: 57) to illustrate how prosody can shape sentence structure.

- (46) a. *Tā bù lái. Wǒ bú qù.*
 he not come I not go
 ‘He doesn’t come. I don’t go.’
 b. *Tā bù lái wǒ bù qù.*
 he not come I not go
 ‘(If) he doesn’t come, (then) I don’t go.’

As Chao pointed out, (46a) consists of two coordinated sentences; however, if no pause is inserted between them, as in (46b), the construction becomes a single sentence—a so-called flowing sentence. Feng (2023)

argues that this syntactic change is essentially the result of a prosodic shift: Two separate SIPs are reanalyzed as a single SIP, leading to the elimination of the pause between the two sentences. According to the “Constraint on Sentential Intonation in Chinese”, within the scope of a single SIP, only one nuclear stress is allowed. That is, only the predicate in the second part, i.e., the verb *qù* in *wǒ bú qù* ‘I won’t go’, bears the nuclear stress (cf. Feng 2003, 2013).

This prosodic merger has a corresponding syntactic consequence: The original coordination of two independent root clauses is reanalyzed as a matrix-subordinate relationship. The first clause becomes some kind of conditional adverbial, thereby losing its root status and its associated nuclear stress. Interestingly, if we explicitly add the conditional conjunctions “if... then...” (*rúguǒ... jiù...*) to both clauses, the two parts of the original flowing sentence are restructured into two relatively independent clauses of a complex sentence, and both are again able to receive their own nuclear stress, as shown in (46):

- (47) *Rúguǒ tā bù lái, wǒ jiù bú qù.*
 if he not come I then not go
 ‘If he doesn’t come, I won’t go.’

In Mandarin, it seems to be a common phenomenon for a conditional sentence—originally composed of two independent root clauses or two relatively independent clauses within a complex sentence—to be reanalyzed as a single sentence via intonational fusion. One such type of “compressive complex sentence” (紧缩句 *jǐnsuō jù*), as discussed by Feng, is also the result of prosodic integration within a conditional construction. Sentences like (48) from Feng, and (49), a classic example from *wúxiá* (武侠, ‘martial heroes’) fiction, are typical examples of this process:

- (48) *Huà bù shuō bù míng.*
 words not say not understand
 ‘If it’s not said, it won’t be understood.’
- (49) *Jiàn zài rén zài, jiàn wáng rén wáng*
 sword exist person exist sword perish person perish
 ‘If the sword is there, the person is there; if the sword perishes, the person perishes.’

Similarly, we can also move from simplicity to complexity, as demonstrated in (47), and reconstruct the two sentences in (48) and (49) from one single SIP into two distinct ones by inserting various syntactic elements, along with a prosodic pause between the two parts. This allows the one SIP to be converted into two full-fledged SIPs.

These examples demonstrate that syntactic complexity or simplicity often parallels the division or merger of prosodic structure. In the case of the barest conditional constructions, it is reasonable to propose that their emergence is driven, at least in part, by prosodic fusion in spoken discourse. The process of simplification may be described in a stepwise fashion as in (50):

(50) Reduction steps towards barest conditionals under prosodic fusion

Starting from a bipartite conditional with two SIPs

Rúguǒ nǐ gāi chī-fàn (pause) nǐ jiù chī-fàn.
 if you should eat.meal PAUS you then eat.meal
 ‘If you should eat the meal, then go eat it.’

Step 1: Omission of C-elements

~~*Rúguǒ nǐ gāi chī-fàn*~~ (pause) ~~*nǐ jiù chī-fàn*~~.

Step 2: Deletion of other non-essential elements, such as subjects and objects

~~*Nǐ gāi chī-fàn*~~ (pause) ~~*nǐ chī-fàn*~~.

Step 3: Omission of the pause marker and completion of the fusion

Gāi chī (pause) *chī*.

Among these simplification steps in (50), only Step 1, which eliminates the C-elements, is syntactically necessary for merging two CPs into one. In contrast, from the perspective of prosody, only Step 3 is obligatory. Step 2 is optional in the sense that it is driven by pragmatic economy. It is also important to note that these steps do not need to occur in a strict sequential order. Interestingly, language data reflect varying degrees of simplification, as seen in the following naturally attested forms: in (51), a conditional clause with only the clause marker omitted, but otherwise fully clausal and prosodically bipartite; in (52), where both clause markers are dropped, but a prosodic boundary in between the two parts remains, which is marked by a pause marker; and in (53), where most

non-essential elements are removed, but the pause remains, resulting in a highly simplified but still prosodically bipartite sentence.

(51) *Gāi chī-fàn, nǐ jiù chī-fàn.*

(52) *Gāi chī-fàn, nǐ chī a.*

(53) *Gāi chī (a), chī!*

These examples from (51) to (53) demonstrate the continuum between full-fledged conditionals and their highly reduced counterparts. However, due to the presence of pause markers between clauses, they should still be analyzed as consisting of two separate SIPs. In other words, it is only when prosodic fusion is complete, i.e., when the pause between the two clauses is eliminated, that the structure can genuinely be considered a single prosodic unit.

Furthermore, we argue that only such truly fused conditionals are subject to the strict constraints on identical pairing as stipulated in the revised PCOB. For instance, in (51) and (52), although the clause structure is bare enough, they do not require all elements to be identical in both parts. This perspective helps explain the counterexamples that Pan & Jiang (2015) raised in response to Cheng & Huang's (1996) original licensing analysis of bare conditionals. Pan & Jiang (2015) point out that in certain bare conditionals, pronouns and *wh*-words appear to be interchangeable, thereby challenging the necessity of strict *wh*-variable pairing. Consider the following example:

(54) *Shéi yào zhè pò chǎng,*
 who want this lousy factory
ràng gěi shéi/tā hǎo le.
 let give who/he a good SFP
 'Whoever wants this lousy factory, let them have it.'

As Cheng & Huang (2020: 179) have observed, in the aforementioned exceptions, the sentences exhibit a different intonation contour from those of canonical bare conditionals. Specifically, a discernible prosodic break is present between the antecedent and consequent clauses, and it is even possible to insert pause particles like *a*.

This observation suggests that a crucial condition for the applicability of the revised PCOB is that the antecedent and consequent clauses must form a unified prosodic unit. That is, PCOB holds only within a single SIP. This provides compelling evidence for syntax–prosody interaction in the licensing of bare conditionals. This analysis of bare conditionals can also be extended to barest conditionals. Specifically, for a conditional to qualify as a barest one, the two original clauses must be merged into a single SIP, with no intervening pause or sentential particles in between them. Once such prosodic unification is achieved, the revised PCOB applies in full force.

It is important to emphasize that this type of fusion occurs exclusively in spoken language, where all sorts of structural reductions happen. As previously discussed, another clear example of the interaction between colloquial register and barest conditional constructions lies in the observation that only highly colloquial modal, typically monosyllabic, can license such constructions. A straightforward rationale behind the interplay of register, prosody, and syntax in these conditionals can be summarized as follows: Within a highly colloquial register, the economy principle drives the prosodic fusion of the antecedent and consequent clauses in conditional sentences. Under the prosodic pressure for these two SIPs to merge into one, superfluous elements (syntactically redundant and semantically non-focused) begin to be omitted. At the same time, syntactic and semantic mechanisms, namely, the PAVQ and the revised PCOB, come into play. The result is the emergence of barest conditionals licensed by highly colloquial modals, characterized by two types of identical pairing.

In sum, the emergence of barest conditionals in Mandarin is driven by prosodic fusion and structural reduction, arising from the interaction between syntactic licensing mechanisms and prosodic constraints. Based upon the prior work on prosodic fusion at the lexical level (e.g., Li & Feng 2015; Feng & Liu 2020), this study presents a novel exploration of fusion at the clausal level.¹⁰

4. CONCLUSION

This paper carries out a thorough investigation into the syntactic and semantic of two types of barest conditionals in Mandarin Chinese. We argue that both types are licensed by a lexical necessity operator, typically

occupying the sentence-initial position. It binds the paired variables in the antecedent and consequent clauses, resulting in the conditional interpretation. Our diagnostic confirms that the variables in barest conditionals are indeed licensed by the unselective binding mechanism.

We identify two variants of the bound variables in barest conditionals, namely, implicit situation variables and *wh*-variables. Crucially, only those variables that serve as information foci are realized at PF as identical pairs, which in turn must satisfy the constraints of the revised PCOB. Specifically, when the focus is on the event itself, the verb phrase containing the situation variable participates in the identical pairing structure. Conversely, when the focus is on the question, it is the *wh*-variable that undergoes identical pairing.

Furthermore, we argue that prosodic fusion of two adjacent clauses in colloquial Mandarin is the driving force behind the emergence of barest conditional structures. The syntactic simplification (from complex to barest) is closely aligned with the prosodic convergence (from bipartite to unified intonational domains). Due to the register constraints of colloquial speech, not all modals are eligible for appearing in barest conditionals: Monosyllabic modals are overwhelmingly favored in colloquial contexts. In sum, the findings of this study reveal an intricate interplay among register, prosody, and syntax in Mandarin. Barest conditionals provide us a perfect testing ground to observe and analyze their interaction.

NOTES

¹ The syntactic and semantic functions of *jiù* in conditional sentences have long been a subject of debate, with no consensus in the literature (see Cheng & Huang 1996, 2020; Hole 2004; Pan & Jiang 2015; Tsai 1999, 2023; Wimmer 2022; Jin & Du 2014; Deng 2018, among others). One basic empirical observation is that when *jiù* is present in a conditional sentence, the strict requirement in bare conditionals for the interrogative word to appear in both the antecedent and the consequent clause in a paired fashion becomes relaxed. For instance, in the bare conditional shown in example (5), if we add *jiù*, the sentence becomes acceptable in a form such as: *Shéi xiān lái, nà ge rén/tā jiù xiān chī* ‘Whoever comes first, that person/he eats first.’ In other words, the interrogative word in the consequent clause can be replaced by another nominal element. We will return to this phenomenon below. Based on this observation, Cheng & Huang (2020: 171) propose that one of the functions of *jiù* is to signal the implicit presence of a conditional operator, namely, an unpronounced *if*, in the antecedent clause. Another basic fact is that, when *jiù* is added, the prosodic boundary between the

antecedent and the consequent appears to be more clearly marked, at least from a prosodic standpoint.

² For further discussion on the interpretive mechanisms of indefinite noun phrases, see Heim (1982, 1990) and Chierchia (1992) on Discourse Representation Theory (DRT). Alternative views have been proposed, for instance, Liu (2016) and Xiang (2020) argue that interrogative elements in conditionals should be analyzed as true interrogatives in terms of semantic type, rather than as indefinites. Li (2021), however, suggests that even if interrogatives in conditionals are treated as *wh*-expressions rather than indefinites, a mechanism akin to Cheng & Huang's unselective binding by a necessity operator can still apply, thereby allowing for a dynamic interpretation of the *wh*-elements in both the antecedent and consequent clauses.

³ Cheng & Huang (1996: 142-143) point out, in *if*-conditionals, the necessity operator binds a situation variable, which explains why *wh*-expression pairing is not required since the prohibition against vacuous quantification is already satisfied in the antecedent. However, they also argue that in *bare* conditionals, the necessity operator simultaneously binds both *wh*-variables and situation variables. This raises an important question: If both situation variables and *wh*-variables are subject to unselective binding by the necessity operator in bare conditionals, why does identical pairing only happen to *wh*-elements rather than to eventual/situational expressions? Cheng & Huang (1996) do not address this issue. Interestingly, in their 2020 paper, they no longer include situation variables as part of the licensing mechanism for bare conditionals. In what follows, this paper proposes a solution to this puzzle based on Chierchia's (1992) analysis of occasion variables.

⁴ Chao (1968: 738) points out that in Modern Chinese, in addition to its verbal use, *ài* also functions as a modal, expressing notions such as willingness or tendency. According to Ōta Tatsuo (1957: 201), the bouletic use of *ài* can be traced back to the Tang Dynasty, as evidenced in poetic lines like Li Bai's *ài tīng sōng fēng qiě gāo wò* (爱听松风且高卧) '(I am) willing to listen to the wind in the pines and lying in repose', where *ài* conveys a sense of preference or desire. Interestingly, the modal usage of *ài* appears to be more widespread, particularly in some other Sinitic languages. For example, in Hakka, *ài* not only expresses volition but can also convey meanings related to prediction or necessity. Similarly, in Southern Min dialects, *ài* may be used to indicate obligation or necessity from the speaker's perspective (see Liu 2000; Zheng 2003).

⁵ In line with Tsai (2025)'s analysis, we argue that this constraint stems from the requirement for prosodic symmetry in barest conditional constructions. Specifically, we observe that the optimal pattern follows a " $\sigma(i) + \sigma(i+1)$ " distribution, where the number of syllables in the second clause is one more than in the first. This ensures a prosodic balance between the initial and final halves of the utterance. When this balance is disrupted—i.e., when the first part is heavier than the second—it results in undesirable prosodic asymmetry.

⁶ Note that, in addition to *ài* and *gāi*, which can directly introduce identical pairings of two *wh*-phrases without accompanying verbs, the other modals listed in (42) still require the presence of verbs that form a constituent with the *wh*-elements. We believe this shows that different syntactic positions are occupied by these modals due to their varying degrees of grammaticalization. We will return to this observation shortly.

⁷ Surely, this does not mean that disyllabic modals are entirely incapable of licensing such constructions. For instance, the disyllabic deontic modal *yīnggāi* can also occur in barest conditionals. However, disyllabic modals such as *yīnggāi* appear to impose restrictions on the prosodic length of the following paired constituent. For instance, the sentence *yīnggāi chī chī* sounds markedly awkward and unnatural, as a contrast, *yīnggāi chīfàn chīfàn* sounds better.

⁸ For a detailed discussion of the grammaticalization of Chinese modals, see Tsai (2010). It is worth noting that the syntactic elevation of modals illustrated in (44) is also regarded by Breitbarth, Delva, and Leuschner (2016) as a pathway of grammaticalization. Another

potential example involving grammaticalization is the use of *néng* ‘can’ in (41). Based on the scope interactions between possibility/necessity modals and negation, Meisterernst (2020, 2025: 107–109) demonstrates that in Late Old Chinese and Early Middle Chinese, possibility modals originally located in the lexical layer, such as *kě* ‘can’ and *néng* ‘be able to’, underwent grammaticalization and syntactic elevation to develop into deontic modals expressing necessity. Due to space constraints and the scope of the present study, we will not explore this topic in further detail here.

⁹ Feng Shengli (2023: 69) proposes that the sentential intonation phrase (SIP) is a prosodic unit at the CP level that is jointly constrained by both prosodic and syntactic features. In sentences without a narrow focus, the sentential intonational stress of the SIP typically coincides with the nuclear stress, and it carries the final boundary intonation of the sentence. This final boundary intonation is characterized by specific phonetic cues that signal the end of a sentence, most notably, the shortening of the final syllable’s rhyme and the lengthening of a silent pause segment. For a detailed discussion of final boundary intonation, see Wang Hongjun (2011).

¹⁰ For another case of clause-level fusion, see Hu (2021: 263–289).

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从句法-韵律界面视角看极简条件句的认可机制

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摘要

本文深入探讨普通话中两类存在动词或者疑问词雷同配对的“极简条件句”：第一类是“该”字条件句，如“该吃吃，该喝喝”；第二类是“疑问词条件句”，如“爱谁谁”（参看蔡维天 2025）。通过对雷同配对的认可机制的分析，本文认为，这两类极简条件句都是由句首模态词认可的条件句。本文主张将这类句子的句首模态词视作显性必要算子，无择约束条件句中的雷同变项，从而赋予条件句全称量化的必要性语义（参看 Cheng & Huang 1996, 2020）。另一方面，雷同变项的种类取决于何者成为信息焦点，而其句首模态词则是认可极简条件用法的关键。在韵律制约的面向上，本文重点探讨了极简条件句的韵律融合现象，提出韵律上的合二为一，即条件句的两个子句融合成为一个单一句调短语，是判断是否为极简条件句和是否需要遵守严格雷同配的最重要指标（参看冯胜利 2017）。本文还进一步揭示了极简条件句所体现出的语体、句法和韵律的互动关系。

关键词

极简条件句、雷同配对、无择约束、韵律融合、句法—韵律界面、模态认可机制